

PUBLISHED: 30-AUG-2017
2015.0 RANGE ROVER (LG), 303-01

ENGINE - TDV8 4.4L DIESEL

RIGHT CAMSHAFT (G1509163)

REMOVAL AND INSTALLATION

12.13.20	CAMSHAFTS - RH BANK - RENEW	LEFT HAND DRIVE, 4400 CC, TDV8, WITHOUT FUEL FIRED BOOSTER HEATER	8.2	USED WITHINS	
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SPECIAL TOOL(S)

303-1239



E81374

303-1239

Adjustment Tool,
Camshaft



E131271

JLR-303-1568

Locking Tool,
Camshaft



E148198

JLR-303-1620

Timing Pin

PART(S)

STEP	REPLACE PART / RENEW PART	PART NAME	
Installation Step 6	Renew Part	Camshaft sprocket bolt	2
Installation Step 8	Renew Part	Timing chain tensioner	1

REMOVAL



NOTES:

- Some variation in the illustrations may occur, but the essential information is always correct.
- Removal steps in this procedure may contain installation details.
- Engine shown removed for clarity.

1. Refer to: [Diesel Fuel System Health and Safety Precautions](#) (100-00 General Information, Description and Operation).

2. Disconnect the battery ground cable.
Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).

3. Refer to: [Cooling Fan](#) (303-03F Engine Cooling - TDV8 4.4L Diesel, Removal and Installation).

4. Refer to: [Right Valve Cover](#) (303-01E Engine - TDV8 4.4L Diesel, Removal and Installation).

5. Refer to: [Generator](#) (414-02B Generator and Regulator - TDV8 4.4L Diesel, Removal and Installation).

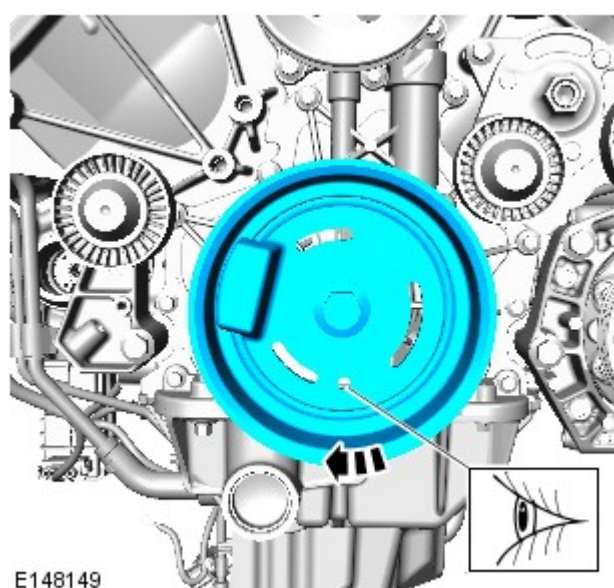
6. Refer to: [Brake Vacuum Pump - TDV8 4.4L Diesel](#) (206-07 Power Brake Actuation, Removal and Installation).

7.

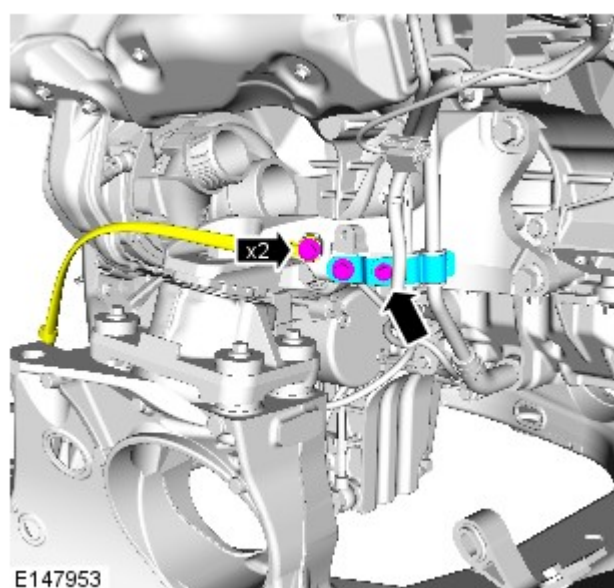


CAUTION:

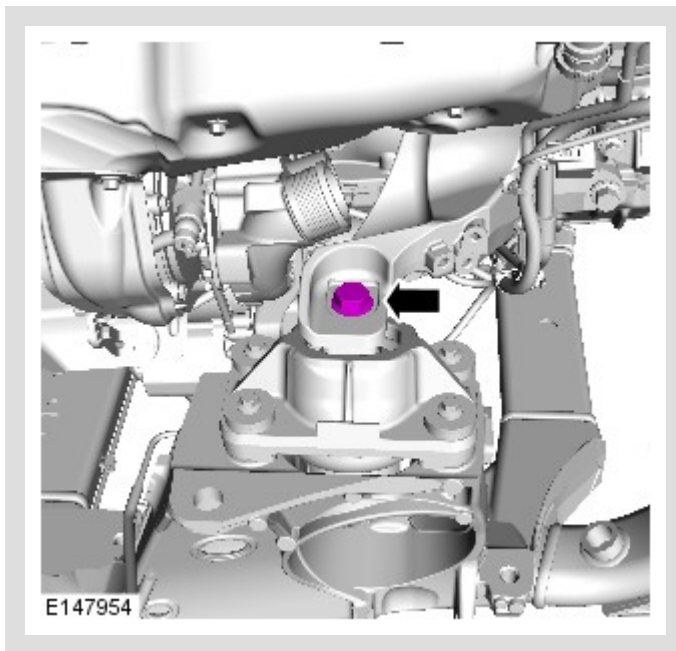
Only rotate the crankshaft clockwise.



8.



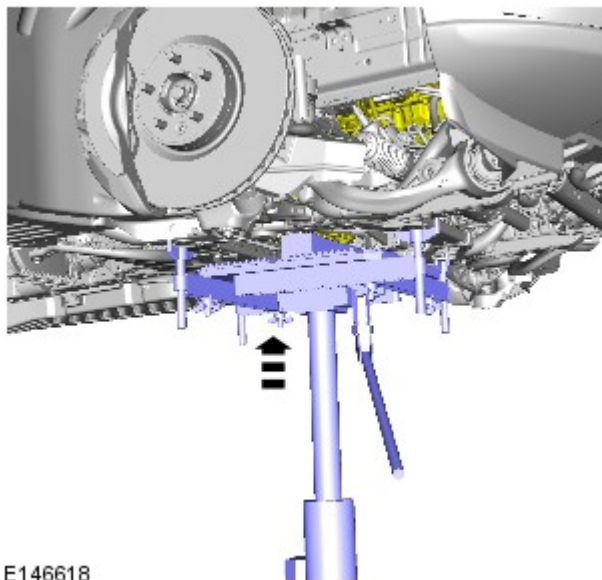
9.



10.

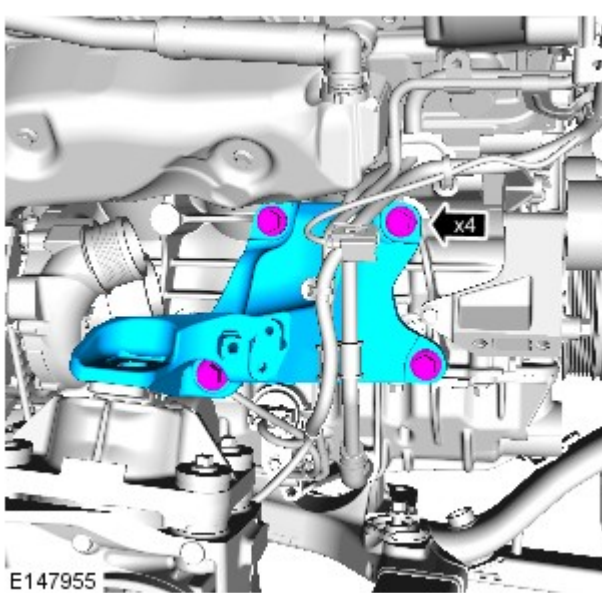
⚠ CAUTION:

Use a wooden block to protect the oil pan when supporting the engine.

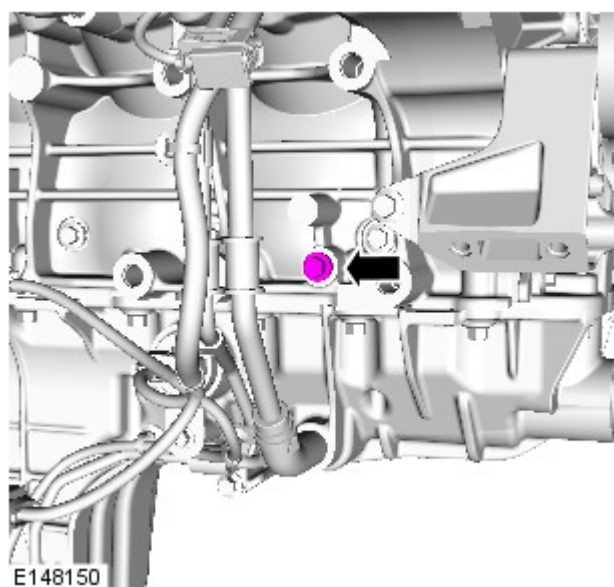


Using a suitable hydraulic jack, raise and support the engine.

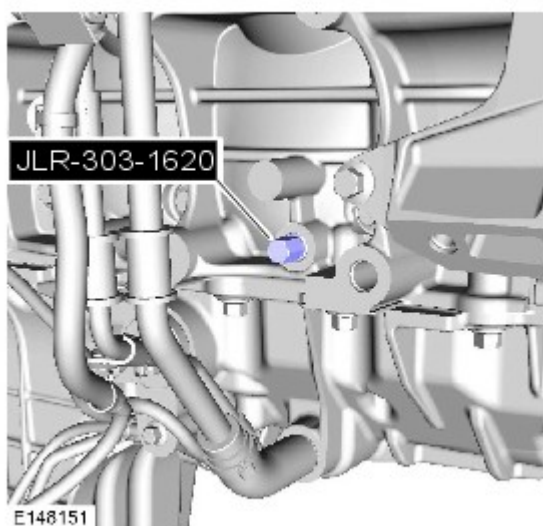
11.



12.

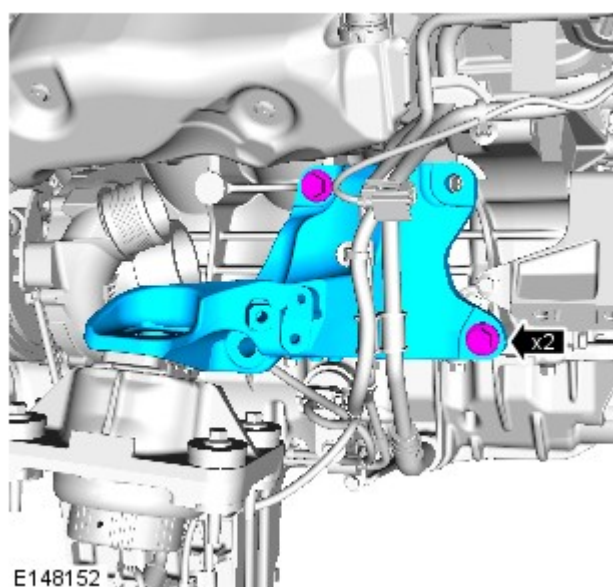


13.

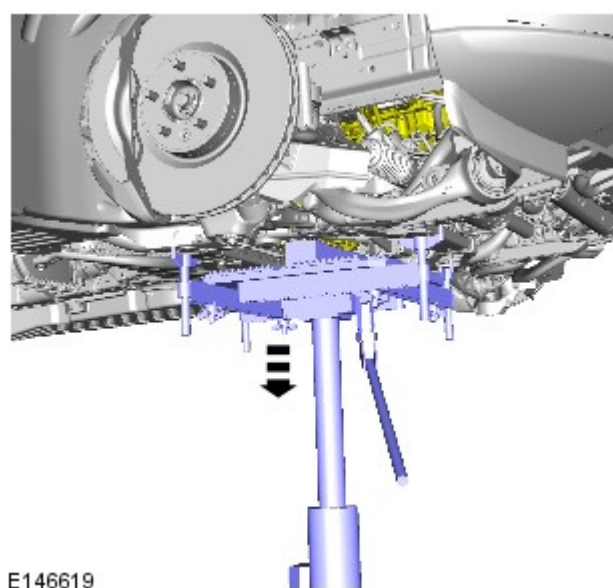


Special Tool(s): [JLR-303-1620](#)

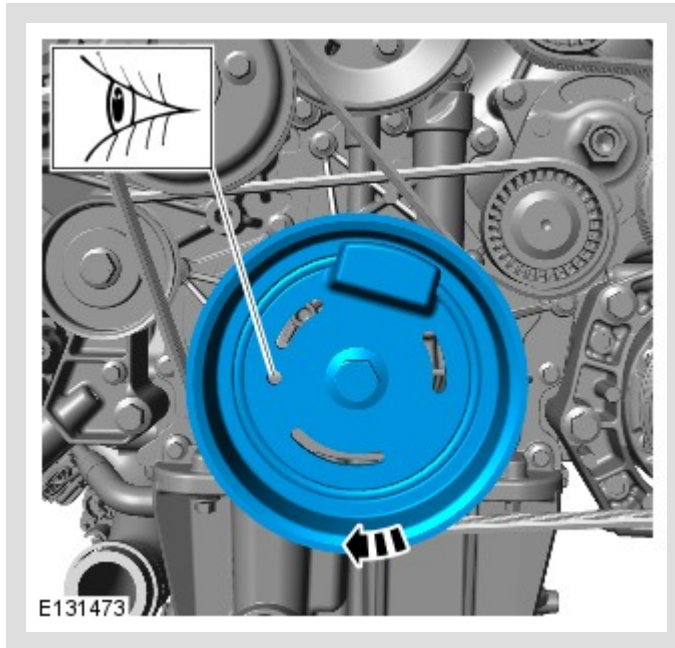
14.



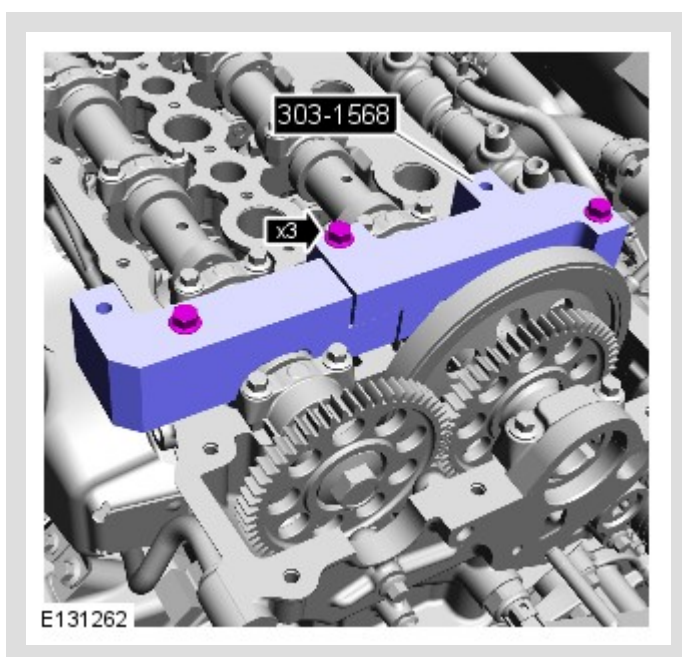
15.



16.



Rotate the engine clockwise until the crankshaft contacts the timing pin. Do not attempt to rotate the engine counter clockwise.

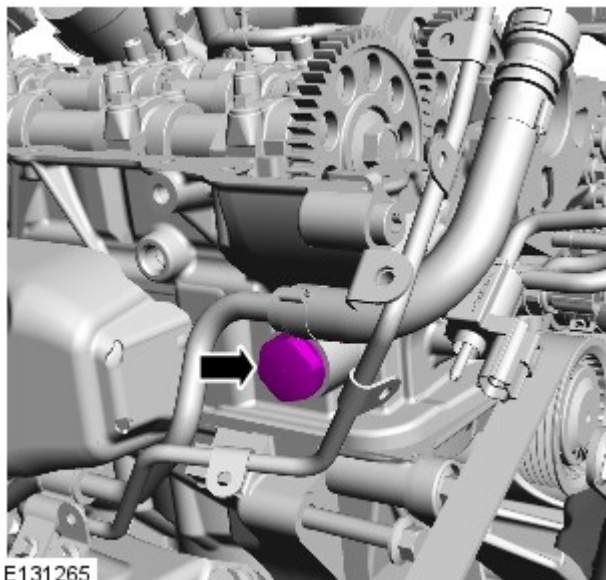


Special Tool(s): [JLR-303-1568](#)

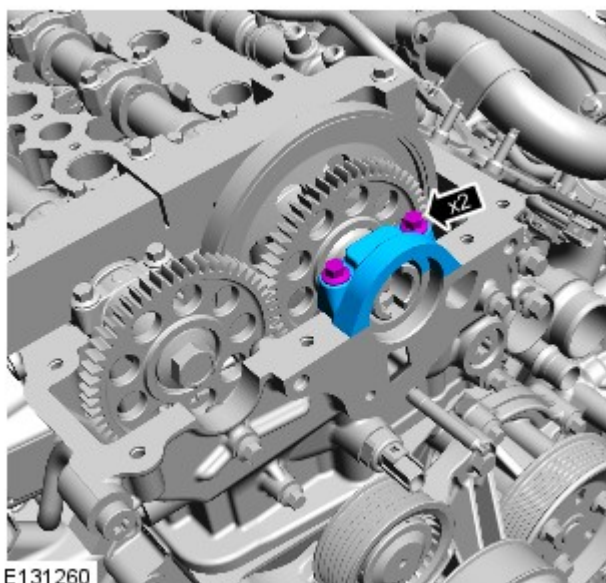
18.

ⓘ CAUTION:

Discard the component.



19.

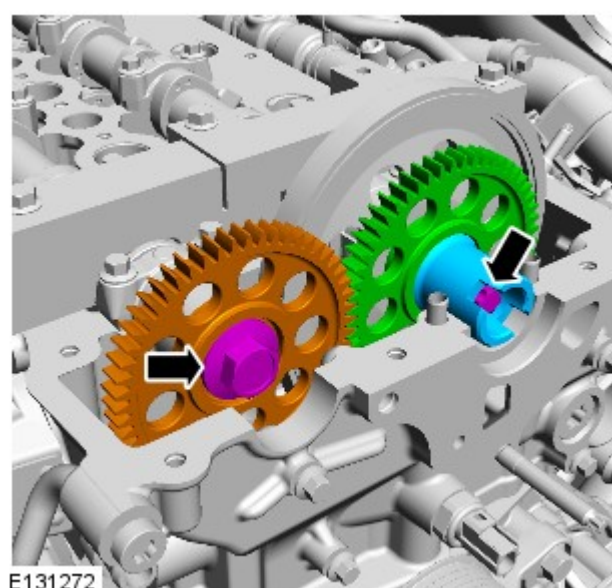


20.

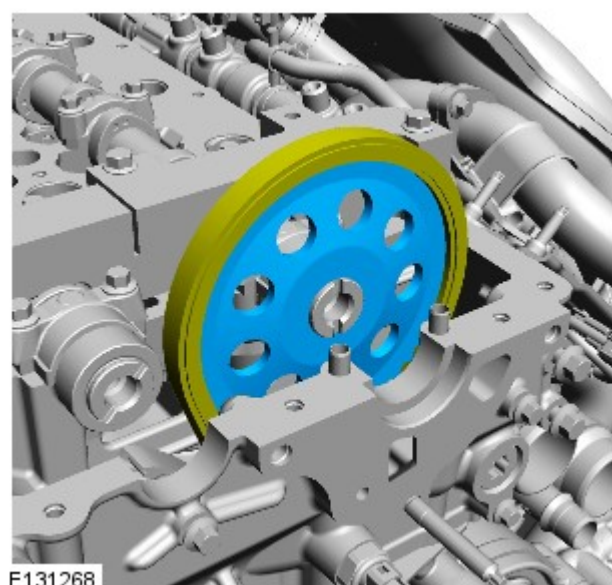


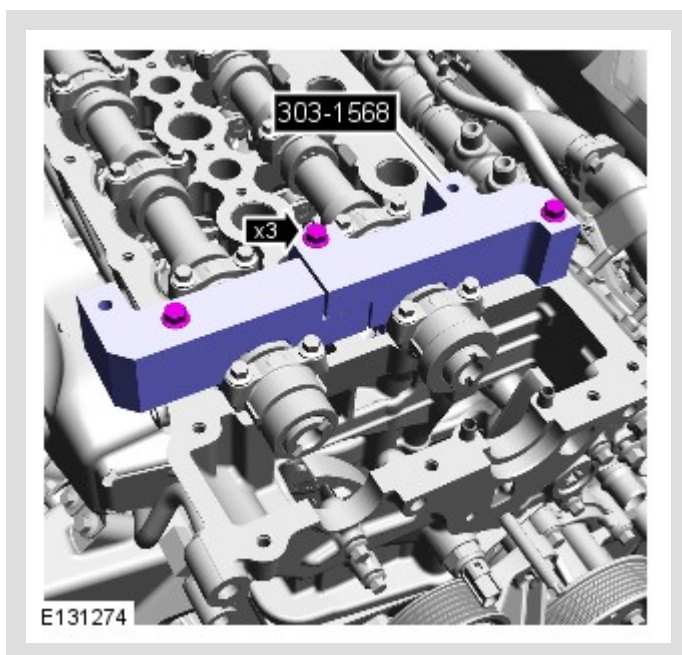
NOTE:

Discard the bolts.



21.



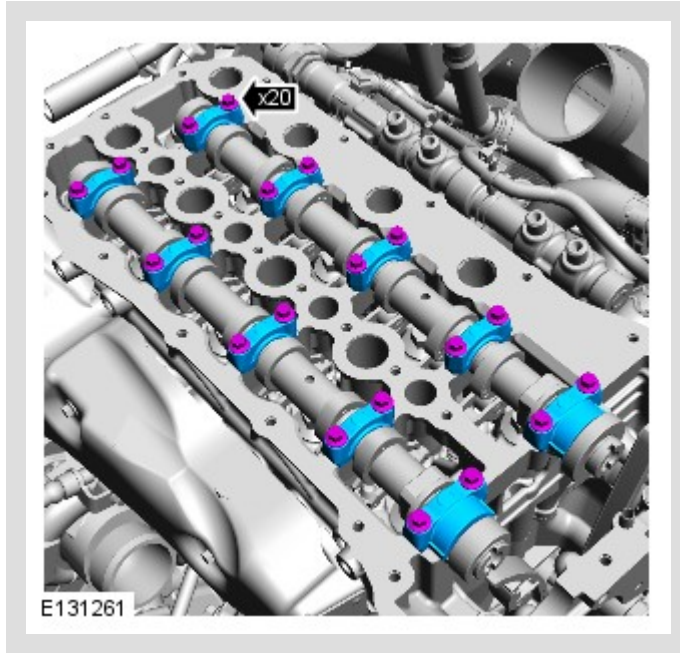


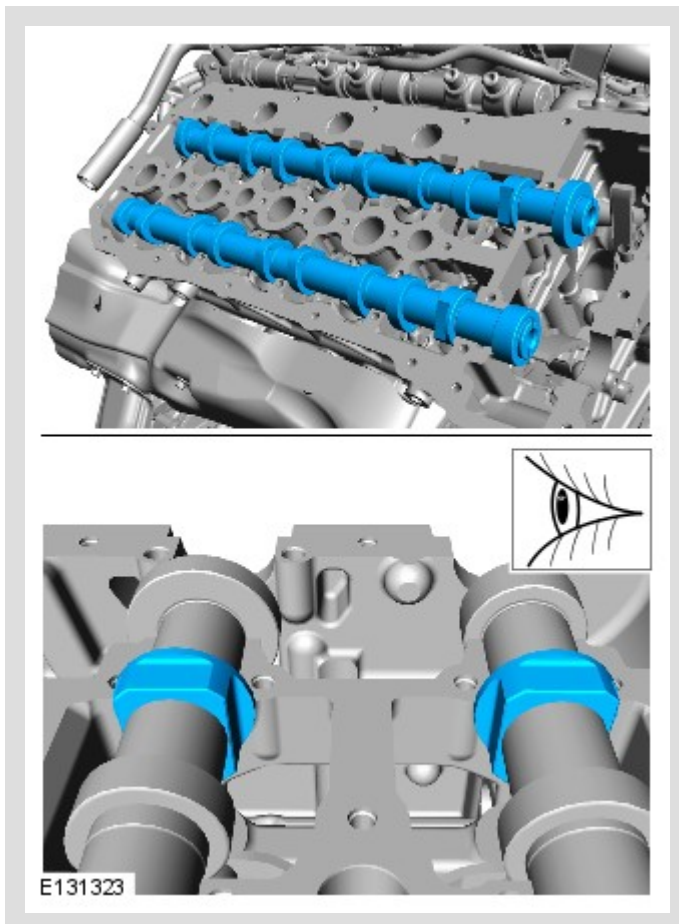
Remove the special tool.

Special Tool(s): [JLR-303-1568](#)

⚠ CAUTION:

Note the fitted position of the component prior to removal.





INSTALLATION

1.



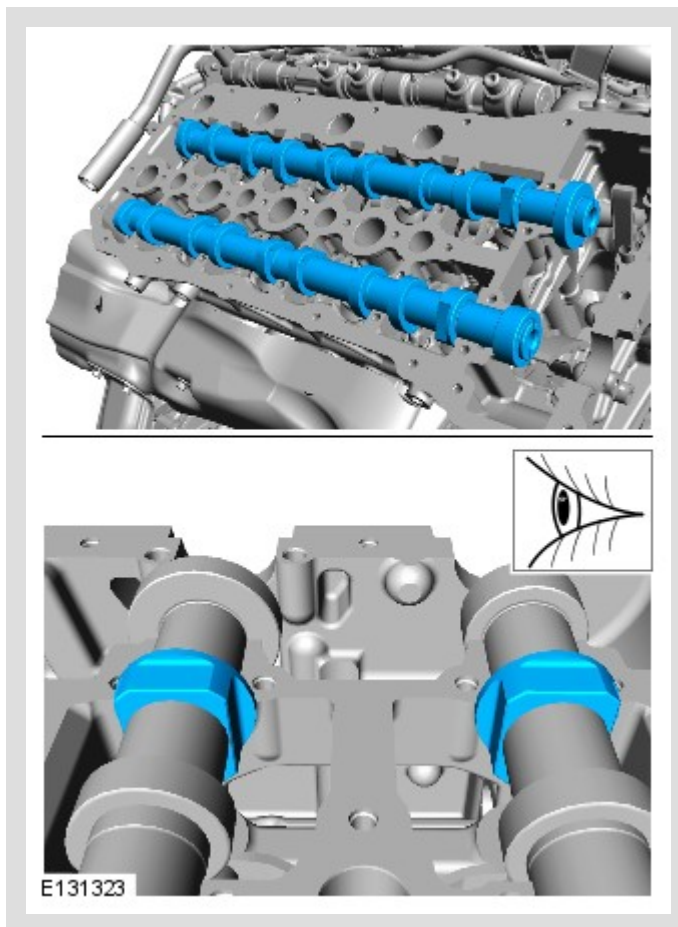
CAUTION:

Make sure that the mating faces are clean and free of foreign material.



NOTES:

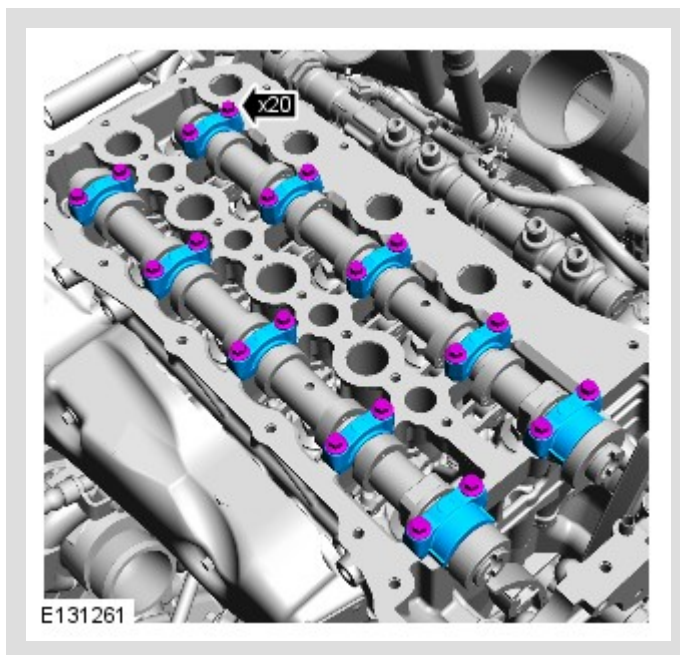
- Lubricate the camshafts and the camshaft bearing caps with oil meeting Land Rover specification prior to installation.
- Align to the orientation noted on removal.



2.

⚠ CAUTIONS:

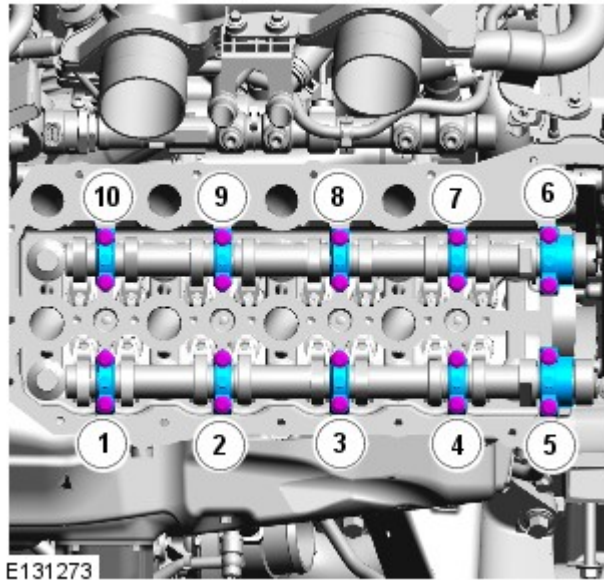
- Make sure that these components are installed to the noted removal position.
- Only tighten the bolts finger-tight at this stage.



3.

⚠ CAUTION:

Tighten the bolts in the sequence shown.



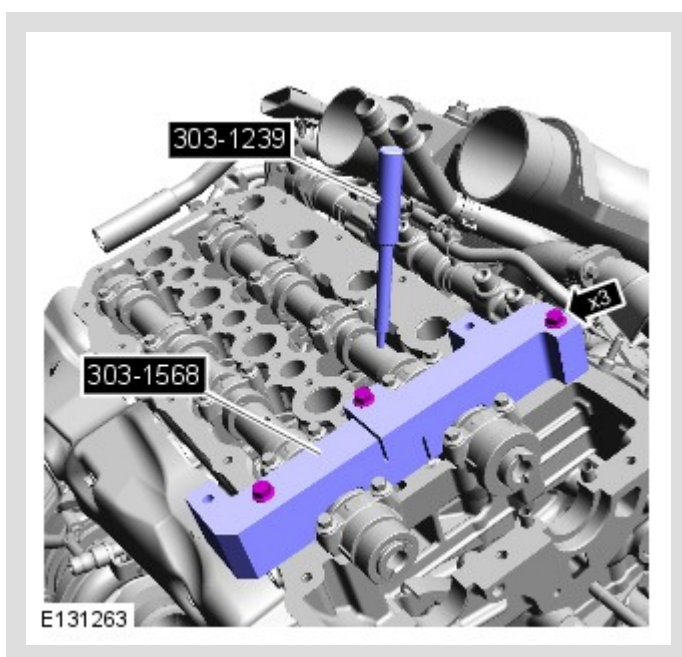
Torque:

Stage 1: Bolts 8,7,9,6,10 5 Nm

Stage 2: Bolts 3,4,2,5,1 5 Nm

Stage 3: Bolts 6,7,8,9,10 10 Nm

Stage 4: Bolts 1,2,3,4,5 10 Nm



- *Special Tool(s):* [JLR-303-1568](#)
- *Special Tool(s):* [303-1239](#)

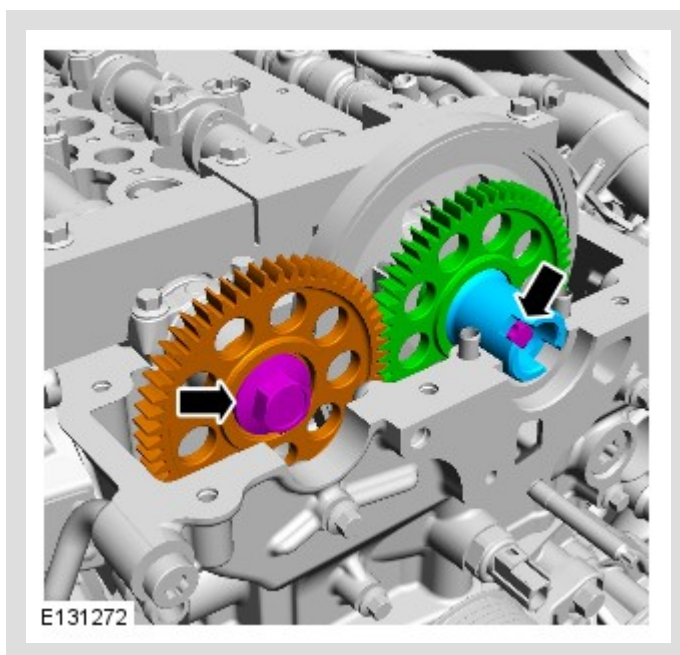
5.



6.

⚠ CAUTIONS:

- Make sure that new bolts are installed.
- Only tighten the bolts finger-tight at this stage.

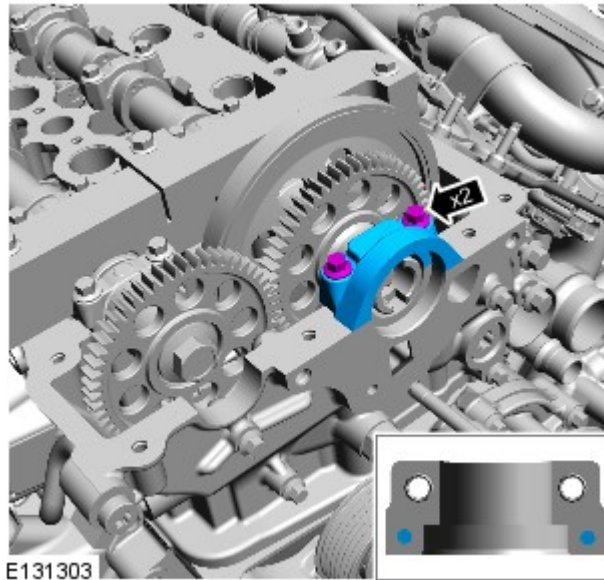


Renew Part: *Camshaft sprocket bolt* : 2 .

7.

⚠ CAUTION:

Make sure that the mating faces are clean and free of foreign material.



Apply a 3.5 mm - 4 mm circle of loctite 518 sealant in the two areas shown.

Torque:

Stage 1: **5 Nm**

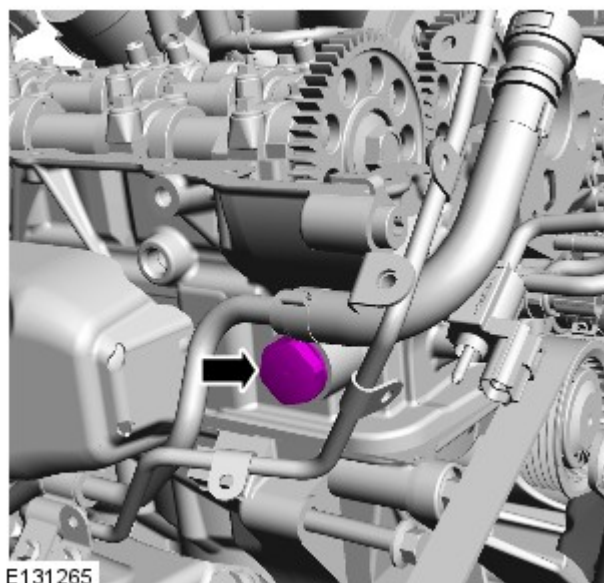
Stage 2: **10 Nm**

8.



CAUTION:

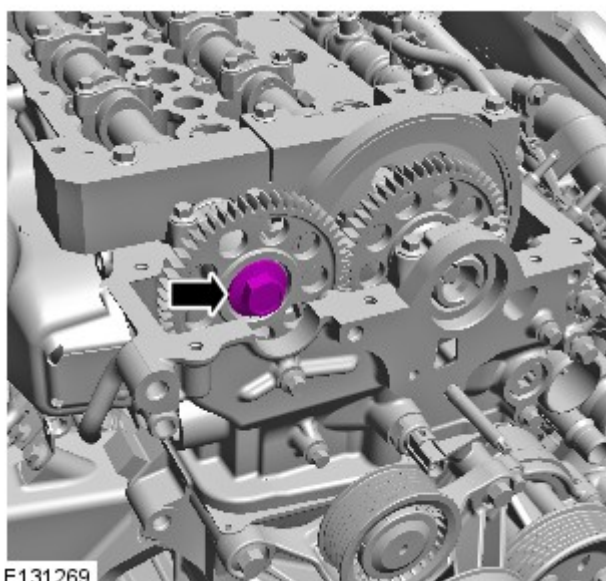
Make sure that a new component is installed.



Renew Part: *Timing chain tensioner : 1* .

Torque: 65 Nm

9.

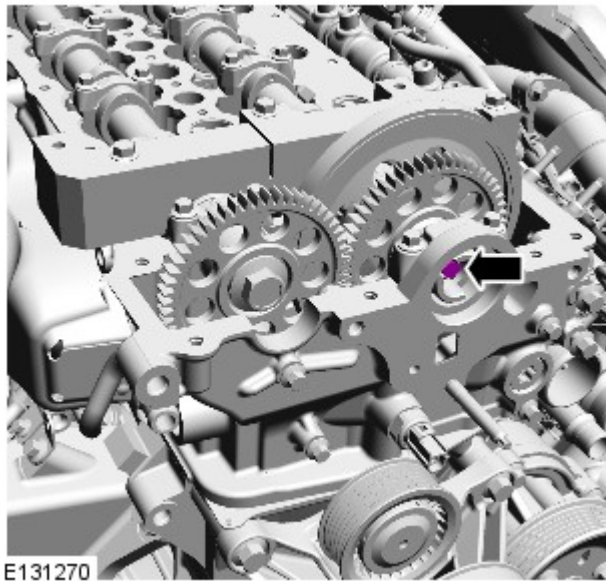


Torque:

Stage 1: 80 Nm

Stage 2: 80°

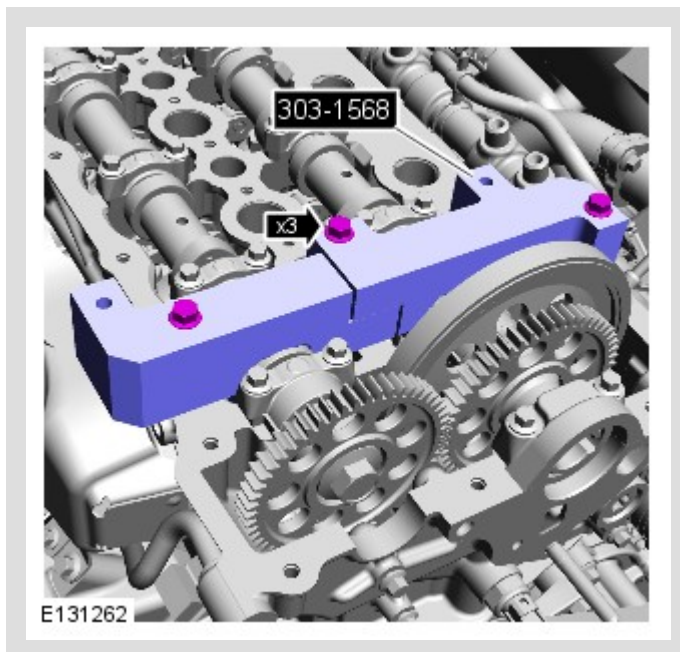
10.



Torque:

Stage 1: 80 Nm

Stage 2: 80°



Remove the special tool.

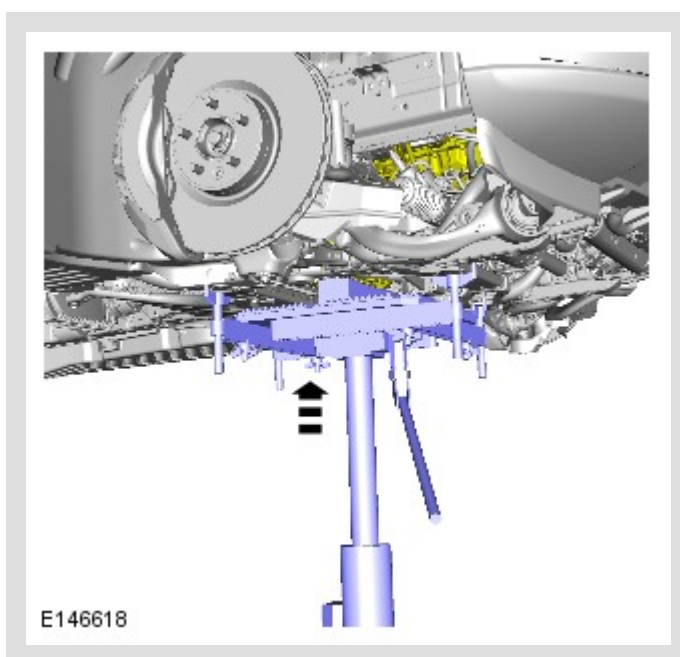
Special Tool(s): [JLR-303-1568](#)

12.



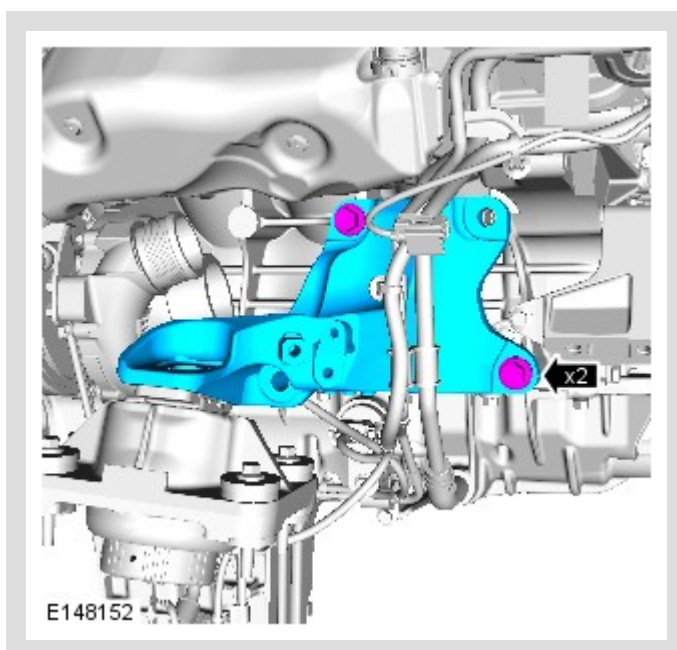
CAUTION:

Use a wooden block to protect the oil pan when supporting the engine.

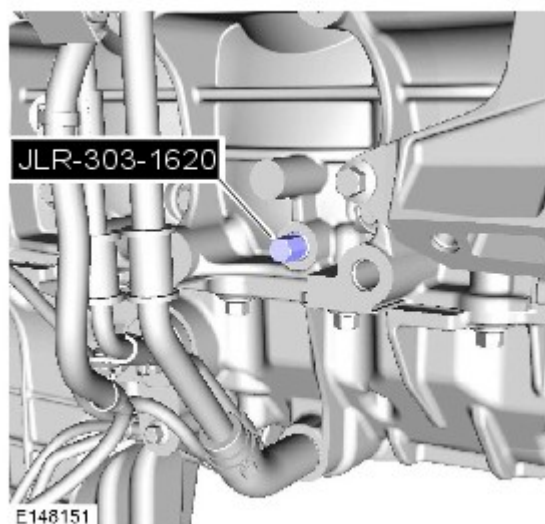


Using a suitable hydraulic jack, raise and support the engine.

13.



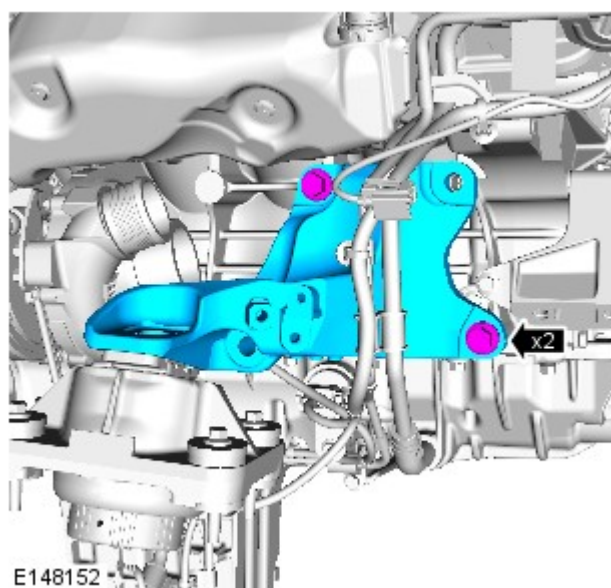
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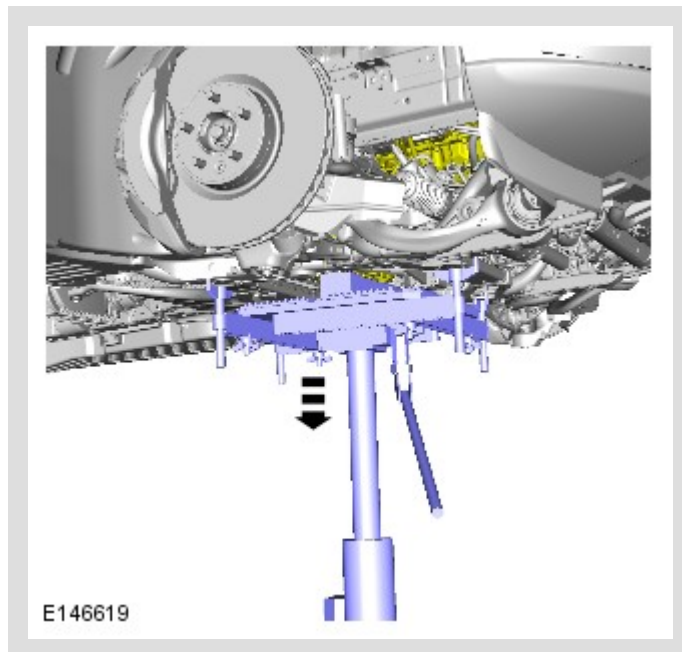
Remove the special tool.

Special Tool(s): [JLR-303-1620](#)

15.



16.

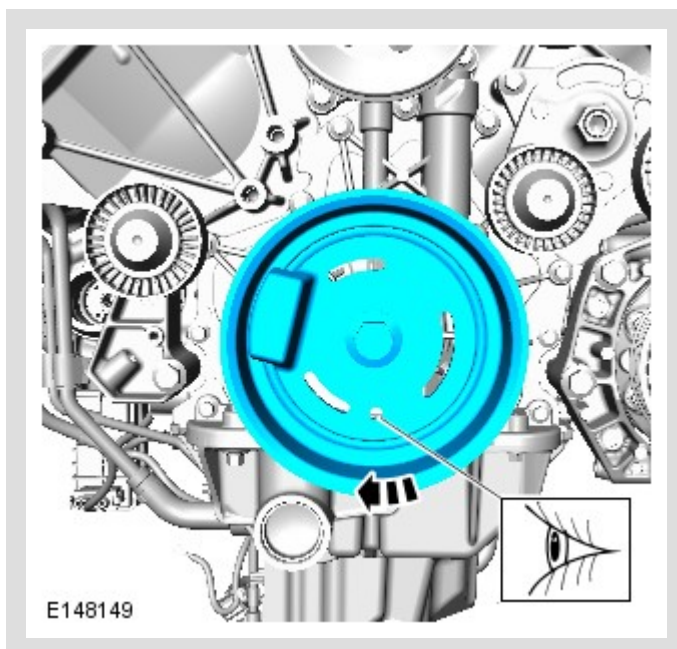


17. Rotate the engine two complete turns clockwise.

18.

⚠ **CAUTION:**

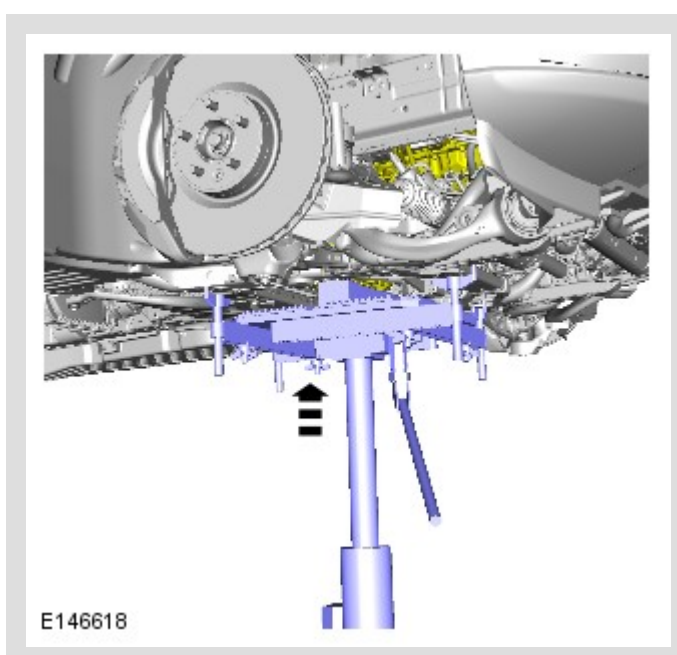
Only rotate the crankshaft clockwise.



19.

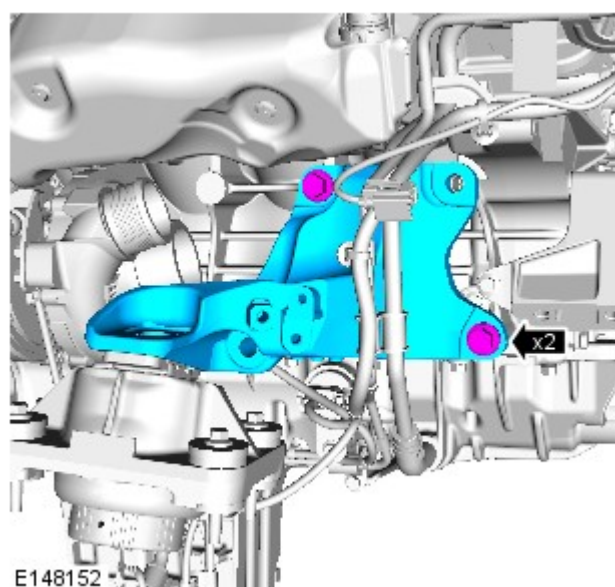
⚠ CAUTION:

Use a wooden block to protect the oil pan when supporting the engine.

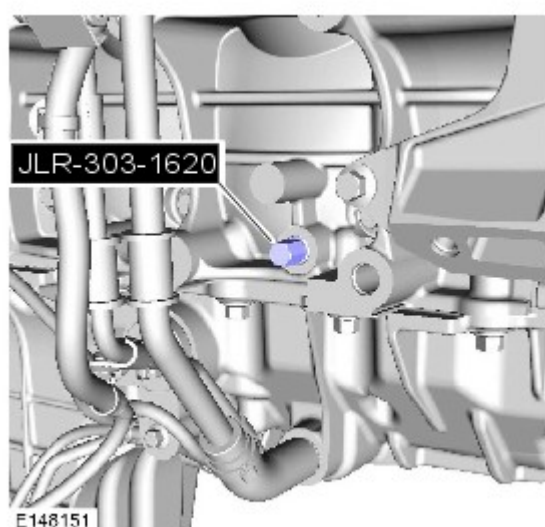


Using a suitable hydraulic jack, raise and support the engine.

20.

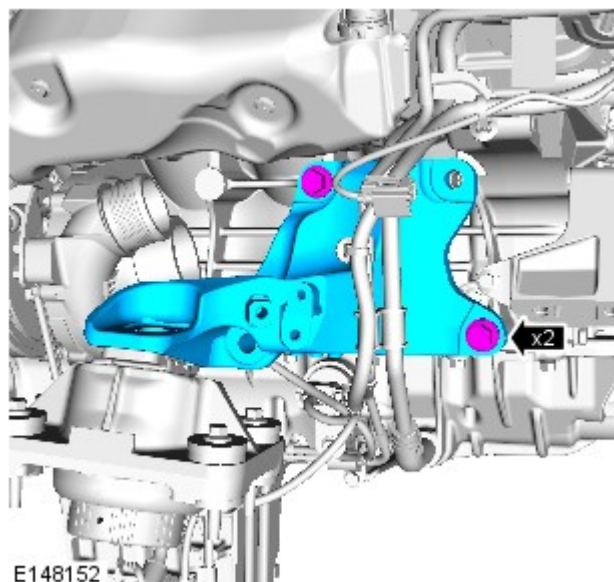


21.

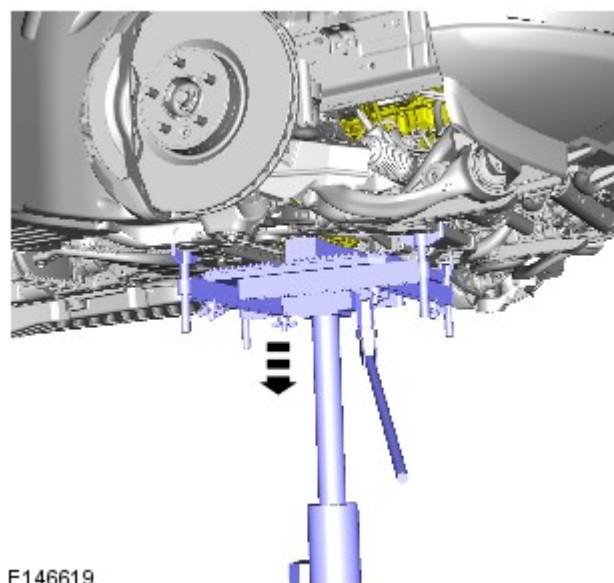


Special Tool(s): [JLR-303-1620](#)

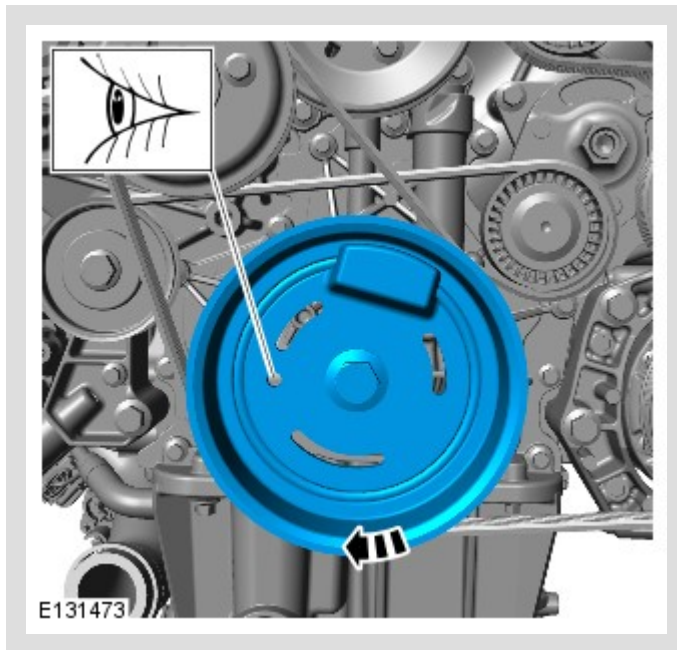
22.



23.



24.



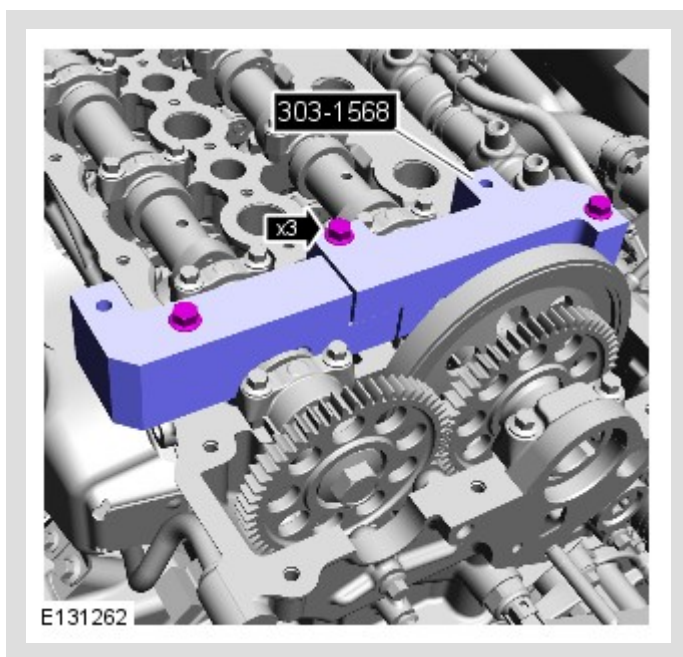
Rotate the engine clockwise until the crankshaft contacts the timing pin. Do not attempt to rotate the engine counter clockwise.

25.



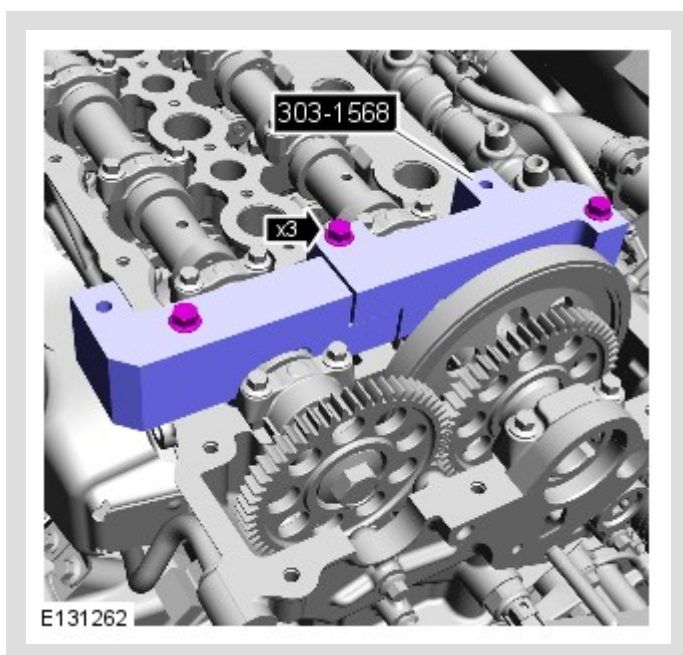
CAUTION:

If the special tool cannot be installed, the timing chain installation steps must be repeated.



Special Tool(s): [JLR-303-1568](#)

26.



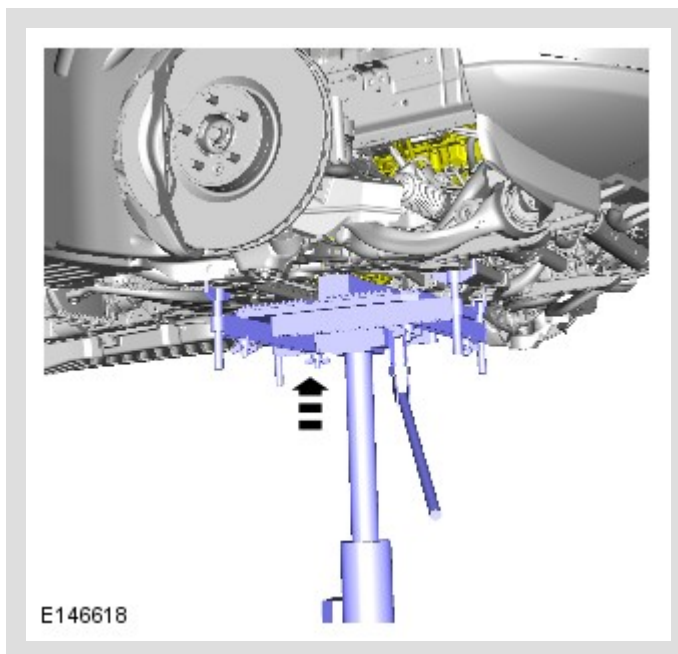
Remove the special tool.

27.

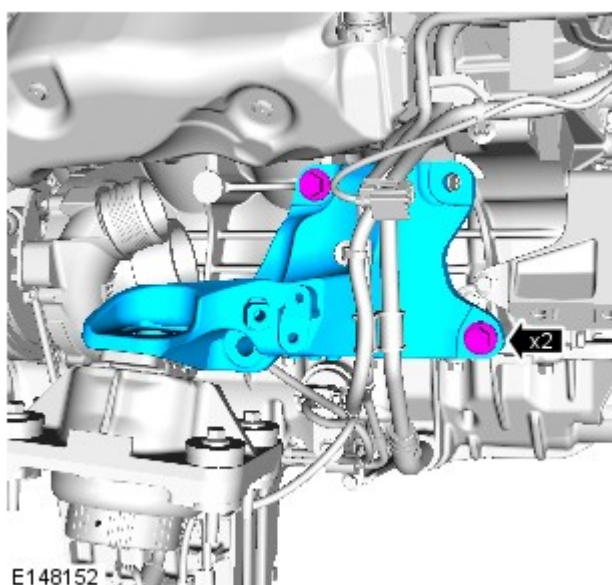


CAUTION:

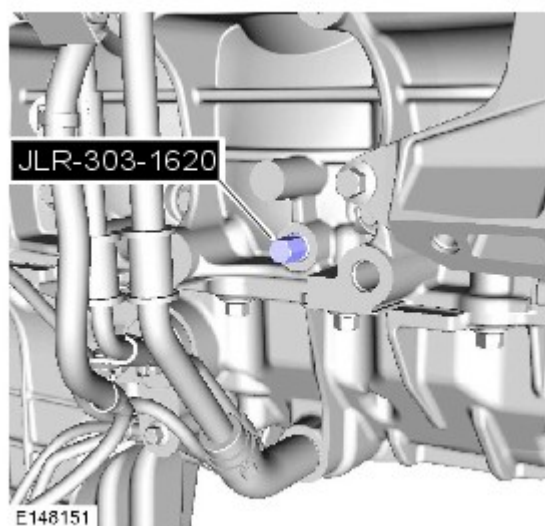
Use a wooden block to protect the oil pan when supporting the engine.



Using a suitable hydraulic jack, raise and support the engine.



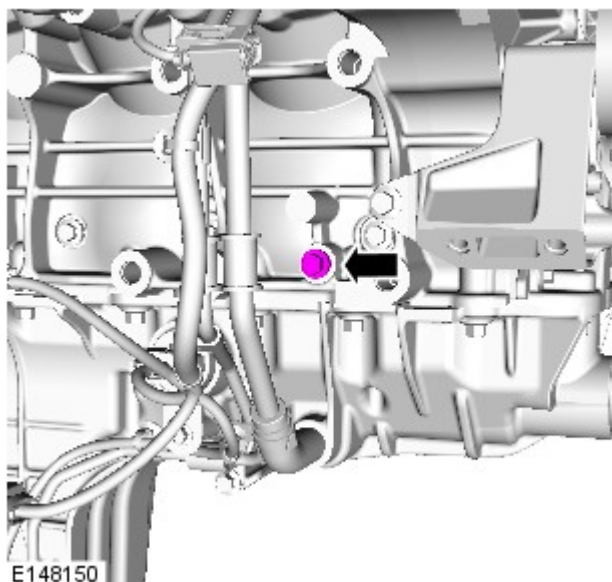
29.



Remove the special tool.

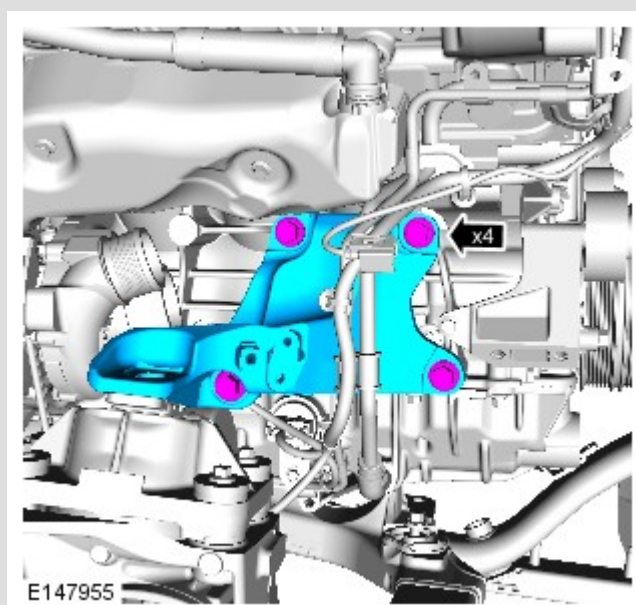
Special Tool(s): [JLR-303-1620](#)

30.



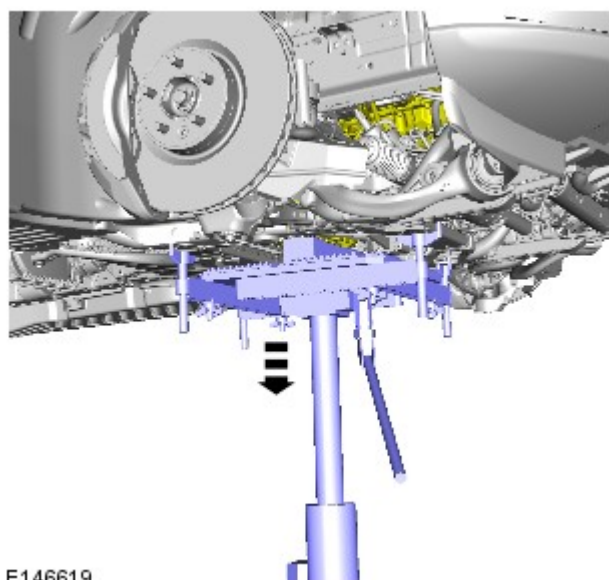
Torque: 23 Nm

31.

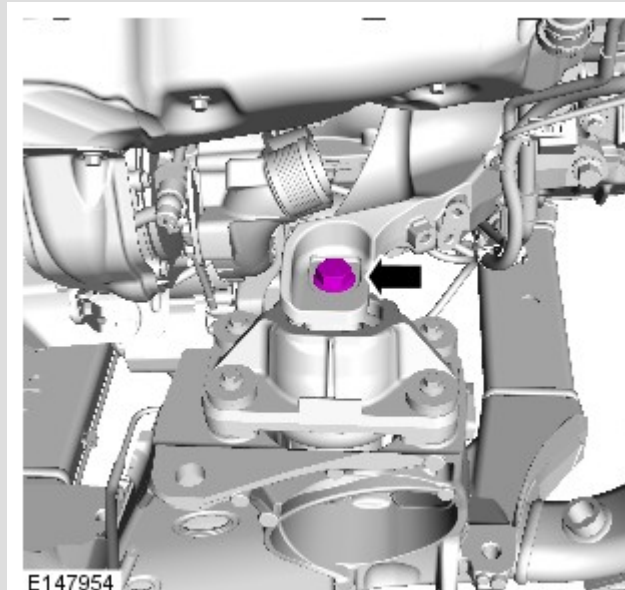


Torque: 119 Nm

32.

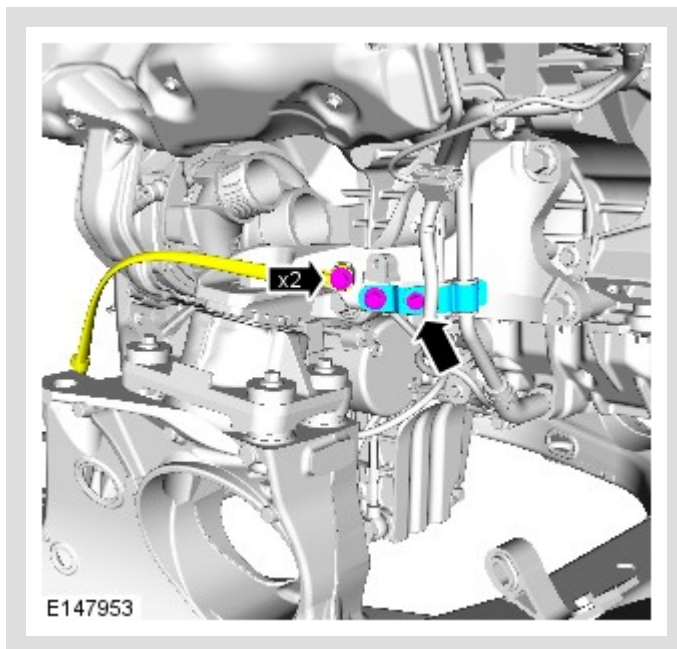


33.



Torque: 110 Nm

34.



Torque:

M8 25 Nm

M6 10 Nm

-
35. Refer to: [Brake Vacuum Pump - TDV8 4.4L Diesel](#) (206-07 Power Brake Actuation, Removal and Installation).

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36. Refer to: [Generator](#) (414-02B Generator and Regulator - TDV8 4.4L Diesel, Removal and Installation).

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37. Refer to: [Right Valve Cover](#) (303-01E Engine - TDV8 4.4L Diesel, Removal and Installation).

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38. Refer to: [Cooling Fan](#) (303-03F Engine Cooling - TDV8 4.4L Diesel, Removal and Installation).

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39. Connect the battery ground cable.
Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).